

7(a)(i)	output voltage / input voltage	B1
7(a)(ii)	no time delay between input and output	B1
	clear reference to <u>change</u> (s) in input and / or output	B1
7(b)(i)	V_{IN} only connected to non-inverting input	B1
	midpoint between R_1 and R_2 only connected to inverting input	B1

Question	Answer	Marks
7(b)(ii)	$\text{gain} = 1 + (R_1 / R_2)$ $25 = 1 + (12 \times 10^3) / R_2$	C1
	$R_2 = 500 \, \Omega$	A1
7(b)(iii)	$V_{\text{MAX}} = 9/25$ $= 0.36 \, \text{V}$	C1
	range is $-0.36 \, \text{V}$ to $+0.36 \, \text{V}$	A1

Question	Answer	Marks
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